

INVERTIS UNIVERSITY, BAREILLY

DEPARTMENT OF APPLIED SCIENCES AND HUMANITIES

LECTURE PLAN

Program Name:	M.Sc. (Physics)	AY	2026-27
Course Name:	MPY 301: Laser Physics	Class / Sem	III Semester
Faculty Name:	Dr. S. K. Jain	Contact	8755169830

No. of Lectures	Topic	Reference	Teaching Method(s)
1	Introduction to laser physics; interaction of radiation with matter: absorption, spontaneous and stimulated emission	T3	C&T, S/P
Unit-1: Properties of Lasers			
2	Einstein's theory: derivation of relation between Einstein A and B coefficients	T3	C&T
1	Condition for light amplification: population inversion	T3	C&T
1	Pumping: mechanisms of producing population inversion	T3, T4	C&T, S/P
2	Two-level, three-level and four-level laser systems	T3	C&T, CHART
1	Optical gain; main components of a laser	T3	C&T, S/P
1	Optical resonator: longitudinal & transverse modes, threshold condition, cavity Q	T3	C&T
1	Properties of laser light: coherence, directionality, monochromaticity, brightness; CW & pulsed lasers	T3	C&T, DEMO
2	Atomic gas laser: construction and working of the He-Ne laser	T1, T3	C&T, S/P, DEMO
Unit-2: Types of Lasers			
1	Ionic gas laser: construction and working of the Argon-ion laser	T1, T3	C&T, S/P
2	Molecular gas laser: construction and working of the CO ₂ laser	T1, T3	C&T, S/P
1	Excimer lasers: principle, construction and applications	T1, T3	C&T, S/P
1	Liquid (dye) lasers: construction and tunability	T3, T4	C&T
1	Solid state lasers: Ruby laser	T1, T3	C&T, S/P
1	Solid state lasers: Nd:YAG laser	T1, T3	C&T, S/P
1	Semiconductor (diode) laser and fibre laser	T3	C&T, S/P
1	Holography: basic principle	T3	C&T
Unit-3: Holography and Fibre Optics			
2	Construction (recording) and reconstruction of a hologram	T3	C&T, S/P, DEMO
1	Holographic Non-Destructive Testing (HNDT); holographic data storage	T3	C&T, SEM
2	Optical fibre principle: total internal reflection; numerical aperture derivation	T2, T3	C&T
1	Types of fibres: step-index/graded-index, single-mode/multimode; fibre losses	T2, T3	C&T, CHART
1	Fibre optic communication: system block diagram	T2, T3	C&T, S/P
1	Fibre amplifiers (EDFA) and fibre-optic sensors	T2, T3	C&T, JAR
1	Saturation spectroscopy (Doppler-free spectroscopy)	T3	C&T
Unit-4: Lasers in Science			
1	Excited state spectroscopy	T3	C&T
1	Nonlinear spectroscopy	T3	C&T, SEM
1	Time domain spectroscopy and its applications	T3	C&T
1	Stimulated Raman emission	T3	C&T

1	Laser fusion	T3	C&T, ET/GL
1	Medical applications of lasers	T3	C&T, SEM
1	Photo-chemical applications	T3	C&T, GD
2	Materials processing: laser drilling and laser cutting	T1, T4	C&T, S/P
Unit-5: Lasers in Industry			
1	Laser welding and laser alloying	T1, T4	C&T, S/P
1	Thermal deposition; Laser Chemical Vapour Deposition (LCVD)	T1, T4	C&T
1	Laser hardening and annealing	T1, T4	C&T
1	LIDAR: principle and applications	T1, T4	C&T, RTCS
1	Case studies/revision: industrial applications of lasers	T1, T4	GD, RTCS
Total Lectures:		44	

Teaching Methods: C&T: Chalk & Talk; S/P: Slides/PPT/Videos; SEM: Seminar; DEMO: Demonstration; CHART: Chart; ET/GL: Expert Talk/Guest Lecture; QUIZ: Quiz; GD: Group Discussion; RTCS: Real Time Case Studies; JAR: Journal Article Review; PD: Poster Design; LAB: Laboratory demonstration.

Text / Reference Books:

- T1 - J. T. Cuxon and D. E. Parker, Industrial Lasers and their Applications, Prentice Hall of India Pvt. Ltd.
- T2 - F. C. Appard, Fiber Optics Handbook, McGraw-Hill.
- T3 - K. Thyagarajan and A. K. Ghatak, Lasers: Theory and Applications.
- T4 - K. Koebner (ed.), Industrial Applications of Lasers, Wiley.